

NUR-SULTAN, Kazakhstan

WMO: 351880

Lat: 51.1703N

Lon: 71.3942E

Elev: 1148

StdP: 14.10

Time Zone: 6.00 (E06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-27.3	-22.1	-34.2	0.8	-27.4	-28.0	1.2	-21.9	32.8	21.5	29.2	21.0	6.4	60	0.650

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.8	90.2	64.0	86.8	63.2	83.7	62.3	67.2	81.6	65.8	80.0	64.6	78.3	8.3	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.8	89.3	70.9	61.1	83.8	70.1	59.4	78.8	69.4	32.4	81.0	31.3	80.0	30.3	78.2	74.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.4	23.5	20.5	-32.7	96.7	5.6	4.3	-36.8	99.7	-40.1	102.2	-43.2	104.6	-47.3	107.7
			-33.1	69.9	5.5	1.9	-37.1	71.3	-40.3	72.4	-43.4	73.4	-47.4	74.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	37.8	2.9	5.4	20.3	43.0	56.2	65.7	67.6	65.2	53.7	40.0	22.7	9.1	(6)
	DBStd	25.77	13.56	12.64	12.10	10.66	8.62	7.11	6.30	7.23	8.60	9.30	12.86	13.05	(7)
	HDD50	6400	1461	1248	920	255	39	2	0	1	57	330	819	1268	(8)
	HDD65	10263	1926	1668	1385	659	290	75	46	88	349	775	1269	1733	(9)
	CDD50	1959	0	0	0	46	232	474	547	472	169	19	0	0	(10)
	CDD65	348	0	0	0	0	18	97	128	94	11	0	0	0	(11)
	CDH74	4615	0	0	0	32	359	1293	1399	1234	293	5	0	0	(12)
	CDH80	1587	0	0	0	4	82	461	494	445	101	0	0	0	(13)
	WSAvg	9.2	9.3	9.8	10.5	9.7	9.6	8.5	8.0	7.8	8.0	8.8	9.7	10.2	(14)
(15) Precipitation	PrecAvg	12.4	0.7	0.6	0.5	0.8	1.3	1.5	2.0	1.5	0.9	1.1	0.8	0.7	(15)
	PrecMax	17.4	2.1	1.6	1.3	2.6	3.3	4.7	6.4	7.0	2.6	2.8	2.1	2.2	(16)
	PrecMin	8.8	0.1	0.1	0.1	0.0	0.2	0.1	0.0	0.0	0.0	0.0	0.1	0.0	(17)
	PrecStd	2.2	0.4	0.3	0.3	0.6	0.7	1.0	1.6	1.3	0.6	0.7	0.5	0.4	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.0	34.4	52.2	78.7	87.8	93.5	94.9	94.0	89.8	73.1	50.3	35.1	(19)
		MCWB	32.8	32.7	43.7	55.9	60.8	64.2	64.9	64.1	62.0	51.6	45.2	33.3	(20)
	2%	DB	30.4	31.2	43.1	71.6	81.9	89.2	89.5	89.3	82.2	66.6	45.0	32.2	(21)
		MCWB	29.5	29.5	38.4	53.1	59.1	63.6	64.8	63.7	58.4	49.9	41.1	31.2	(22)
	5%	DB	26.7	28.3	38.7	66.8	78.1	85.6	85.7	84.6	75.9	61.4	41.0	29.8	(23)
		MCWB	25.6	27.1	35.7	50.8	57.3	62.5	64.3	62.7	56.3	48.1	38.5	28.5	(24)
	10%	DB	23.1	24.7	35.5	61.4	73.9	81.9	81.7	80.9	70.7	55.7	37.4	26.8	(25)
		MCWB	22.0	23.4	33.4	48.2	55.9	61.8	63.6	61.8	54.3	45.4	35.0	25.7	(26)
	0.4%	WB	32.9	33.0	45.0	57.3	64.5	68.2	69.2	68.4	63.2	54.3	46.8	33.5	(27)
		MCDB	33.7	34.3	50.6	73.3	79.2	83.8	84.1	83.6	83.7	67.2	48.7	34.0	(28)
(29) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	29.6	29.5	38.9	54.5	61.2	66.4	67.5	66.1	59.9	51.4	41.7	31.3	(29)
		MCDB	30.3	30.6	42.2	68.2	76.4	82.3	80.8	81.0	77.1	63.2	44.2	32.2	(30)
	5%	WB	25.6	26.8	35.8	51.9	59.2	64.8	66.3	64.5	57.6	49.0	38.1	28.4	(31)
		MCDB	26.4	28.0	38.2	64.4	73.5	79.8	79.5	79.7	72.6	59.4	40.4	29.2	(32)
	10%	WB	22.0	23.4	33.4	49.4	57.4	63.1	65.0	63.0	55.3	46.3	34.9	25.7	(33)
		MCDB	23.0	24.6	35.1	60.2	71.5	77.4	77.7	77.9	69.1	54.7	37.0	26.7	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	17.5	19.0	17.7	21.7	24.4	23.6	21.8	23.9	24.7	19.6	14.3	16.1	(35)
		MCDBR	13.8	16.7	17.0	30.9	30.6	29.9	28.8	30.9	34.3	30.5	16.1	15.1	(36)
		MCWBR	13.4	15.9	12.8	15.1	13.6	11.5	10.8	12.0	15.7	17.3	12.5	14.4	(37)
	5% WB	MCDBR	14.0	15.7	15.9	27.7	26.4	25.5	23.8	26.6	30.8	26.9	14.6	14.2	(38)
		MCWBR	13.8	15.3	12.4	14.6	13.4	11.3	10.1	11.6	15.0	16.2	12.0	13.8	(39)
(40) Clear-Sky Solar Irradiance	taub		0.293	0.325	0.375	0.410	0.404	0.410	0.430	0.409	0.359	0.342	0.327	0.297	(40)
	taud		2.056	1.998	1.970	2.133	2.219	2.267	2.214	2.253	2.346	2.301	2.146	2.048	(41)
	Ebn at Noon		218	243	253	262	270	268	260	259	262	240	203	192	(42)
	Edh at Noon		26	36	46	44	42	41	42	39	31	27	24	23	(43)
(44) All-Sky Solar Radiation	RadAvg		443	800	1222	1523	1900	2019	1855	1650	1236	681	327	298	(44)
	RadStd		42	33	117	131	162	157	129	117	86	92	36	42	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page